MIDTERM PROJECT • DATA SCIENCE

Mental Health of University Students in the Digital Age

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Project Overview

This project investigates how daily social media usage shapes the mental health and academic performance of university students. Real-world student datasets are analyzed to move beyond assumptions and let the data reveal where the relationship is strongest.

RESEARCH QUESTION

"How does daily social media usage influence the mental health and academic performance of university students?"

Variables Under Investigation

- **Social Media Hours/Day**
- **GPA**
- **Stress Level**
- **Sleep Hours/Day**
- **Study Hours/Day**

Dataset

Four complementary student datasets ground each research question in real data

Depression Student Dataset

Source: Kaggle — "Depression Student Dataset"
Size: 502 students × 11 variables

Key variables: Age, Gender, Academic Pressure, Study Satisfaction, Sleep Duration, Study Hours, Financial Stress, Suicidal Thoughts, Depression (target)

Why it fits: *Links sleep, stress, and academic pressure to a clinically framed depression outcome.*

Student Lifestyle Dataset

Source: Kaggle — "Student Lifestyle Dataset"
Size: 2,000 students × 8 variables

Key variables: Study Hours/Day, Extracurricular Hours, Sleep Hours, Social Hours, Physical Activity Hours, GPA, Stress Level (target)

Why it fits: *Quantifies time-tradeoffs (sleep vs. social/screen hours vs. study) that drive GPA and stress.*

Mental Health Student Dataset

Source: Kaggle — MentalHealth_StudentDataset.xlsx
Size: 500 students × 21 variables

Key variables: Daily_SM_Hours, Sleep_Hours, Anxiety_Score, Stress_Score, Depression_Score, GPA, Wellbeing_Score, Sought_Mental_Health_Support

Why it fits: *Strongest signal in the project: SM hours correlate r=0.85 with anxiety, r=−0.83 with wellbeing.*

Social Media Impact Dataset

Source: Kaggle — Social_media_impact_on_life.csv
Size: 1,705 records × 11 variables (1,000 students)

Key variables: Avg_Daily_Usage_Hours, Most_Used_Platform, Affects_Academic_Performance, Sleep_Hours_Per_Night, Mental_Health_Score, Overall_Impact

Why it fits: *Large-sample validation: usage hours correlate r=−0.83 with mental health score, r=−0.82 with sleep.*

Data Preparation

All four datasets arrived clean — preprocessing focused on transformation and feature engineering

1 Missing Values

Checked all columns in all four datasets with isnull().sum(). Result: 0 missing values across 502 + 2,000 + 500 + 1,705 records — no imputation needed.

2 Duplicate Records

duplicated().sum() found 0 duplicate rows across all four datasets; drop_duplicates() applied defensively before analysis in every notebook.

3 Feature Engineering

Lifestyle data: created Total_Productive_Hours and Sleep_Category (Poor/Normal/Good). Both new SM datasets: binned hours into SM_Usage_Level (Low/Medium/High).

4 Derived Categories

Binned continuous GPA into GPA_Category (Low / Medium / High) to test associations with Stress_Level via chi-square.

5 Type & Structure Checks

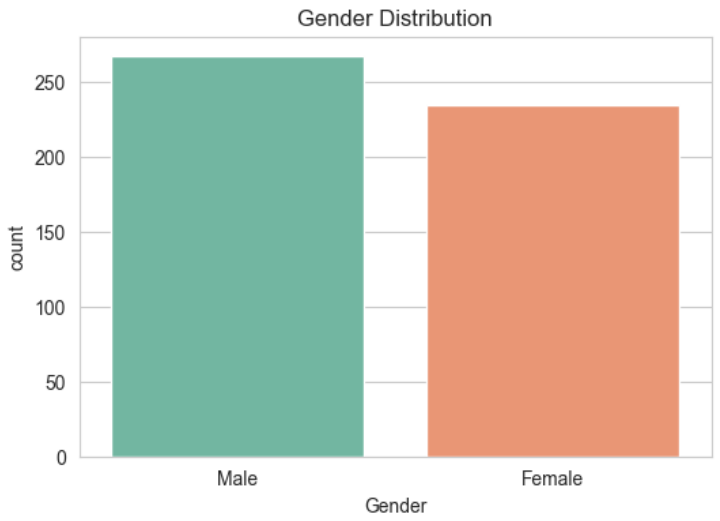
df.info() / describe() confirmed correct dtypes throughout. Social Media Impact data: Student_ID repeated across rows, so a unique Record_ID was added as primary key.

6 Outlier Review

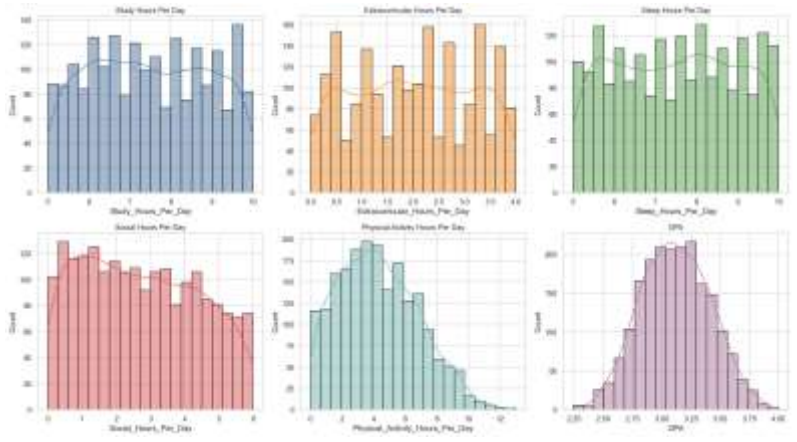
Boxplots across all four datasets (incl. Daily_SM_Hours, Anxiety_Score, Mental_Health_Score) showed no impossible values; extreme points were retained as valid high-distress cases.

Exploratory Data Analysis

Depression Dataset



Gender split: 267 male / 235 female — a fairly balanced sample.



Ages cluster 18–31 (mean 26.2); most lifestyle hours are broadly distributed with no missing data.

502

students surveyed

0

missing values & duplicate rows found

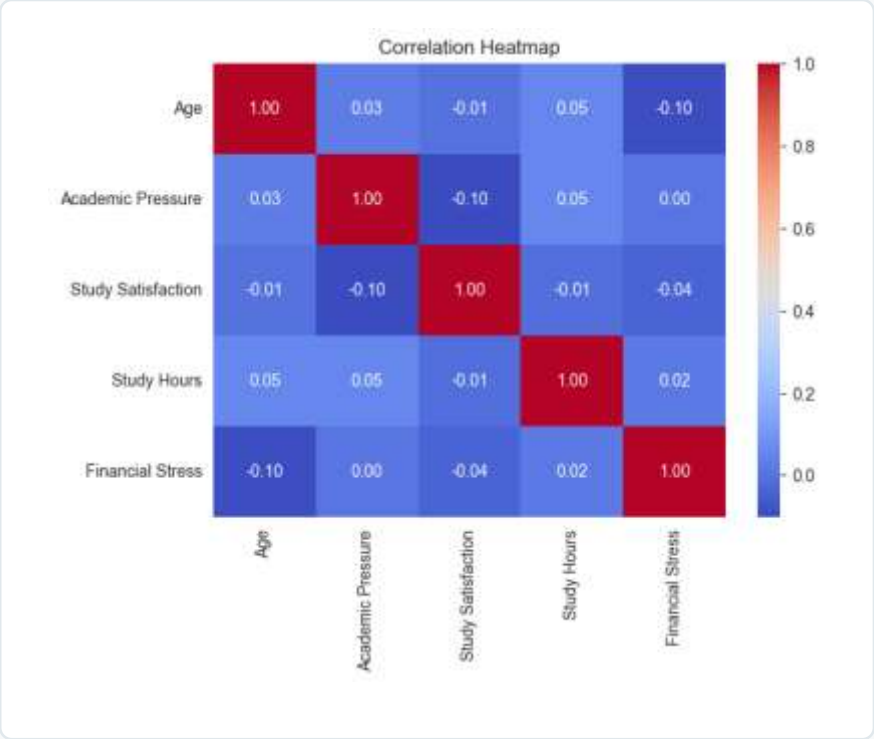
58.7%

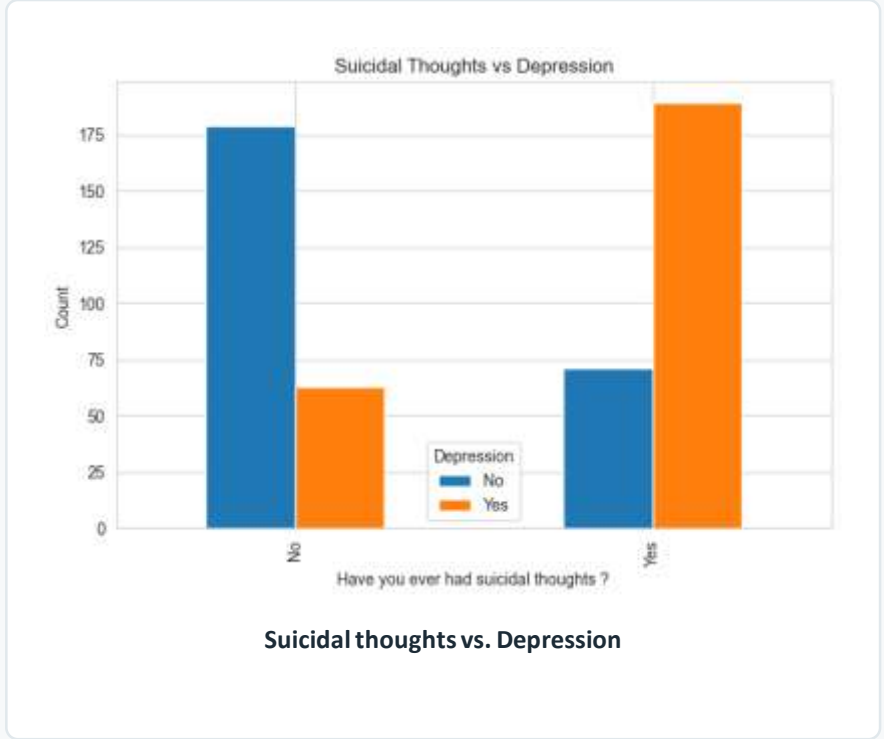
of students report a Depression diagnosis label = “Yes”

Data-quality check: no missing values and zero duplicate rows across all 11 columns — the dataset required no cleaning, only transformation for analysis.

Exploratory Data Analysis

Depression Dataset — Relationships & Significance Tests





Suicidal thoughts vs. Depression

Chi-square test

$p < 0.001$

Suicidal thoughts are significantly associated with Depression (χ^2 = 107.3)

T-test

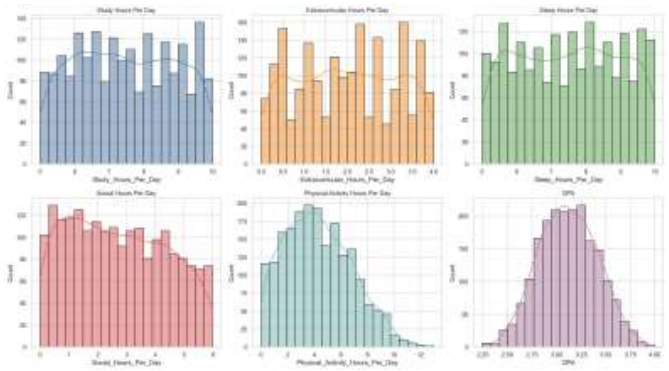
$p < 0.001$

Depressed students study ~1.5 hrs/day more on average (7.2 vs 5.7 hrs) — pressure, not idleness

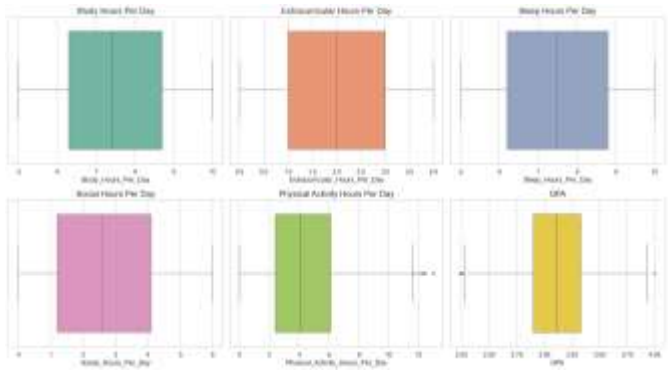
Depressed students also report higher Academic Pressure (3.66 vs 2.34) and Financial Stress (3.35 vs 2.50), and lower Study Satisfaction (2.67 vs 3.48).

Exploratory Data Analysis

Lifestyle Dataset — Distributions & Outlier Check



Daily-hour variables (study, sleep, social, activity, GPA) are roughly symmetric with no extreme skew.



Boxplots confirm no impossible values (e.g. negative hours, GPA outside 0–4.0).

Summary Statistics (2,000 students)

Study Hours/Day: mean 7.48, range 5–10 • **Sleep Hours/Day:** mean ~7.5 • **GPA:** range 2.2–4.0

No column required missing-value imputation, and every numeric field fell within a physically plausible range — confirming the dataset was clean at the source and ready for feature engineering.

2,000

student records, 8 raw variables

0

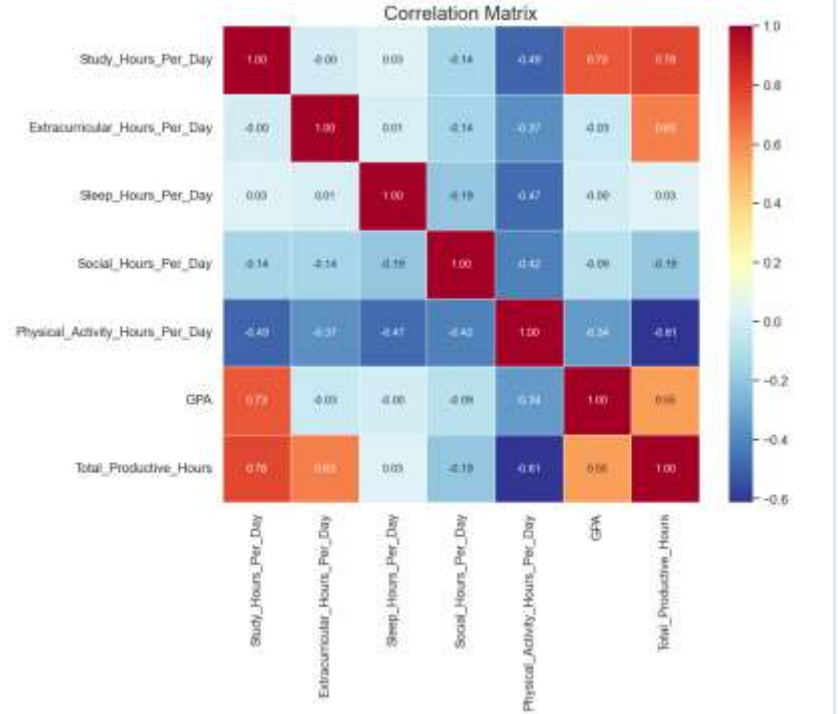
missing values & duplicate rows

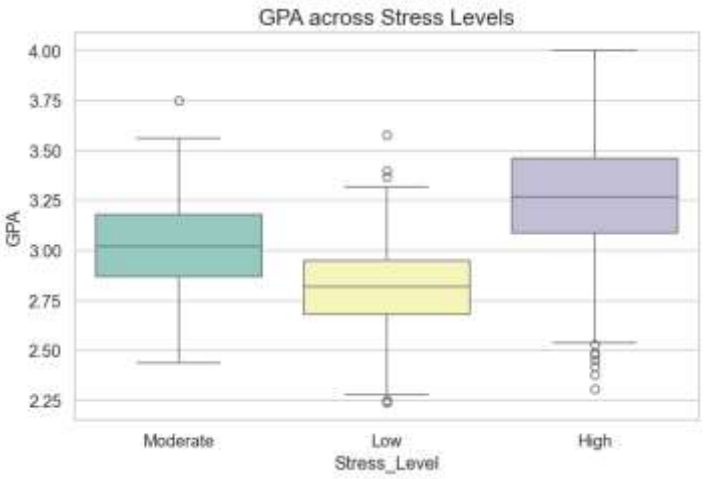
Boxplots

confirm no impossible values (e.g. negative hours, GPA outside 0–4.0) and only mild, expected spread — no outlier removal required

Exploratory Data Analysis

Lifestyle Dataset — What Actually Drives GPA & Stress





GPA falls sharply as stress rises

Correlation with GPA

Study Hours/Day	**+0.73**
Physical Activity	**−0.34**
Social Hours	**−0.09**
Sleep Hours	**−0.00 (n.s.)**

ANOVA: GPA by Stress

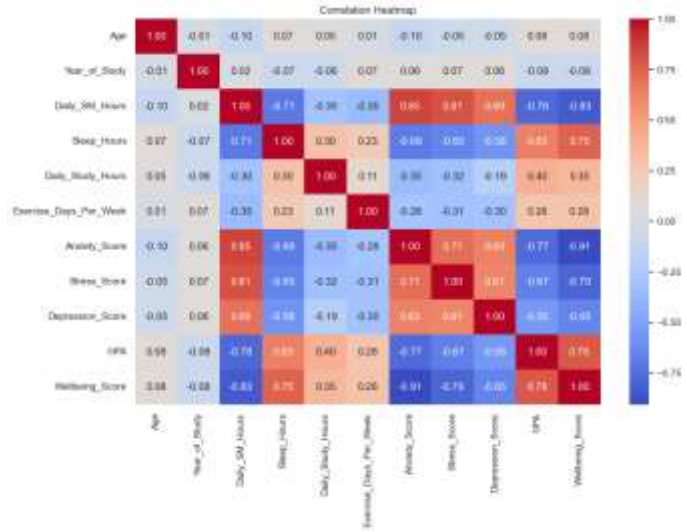
$p < 0.001$

(F = 434.9) — GPA differs significantly across Low / Moderate / High stress groups

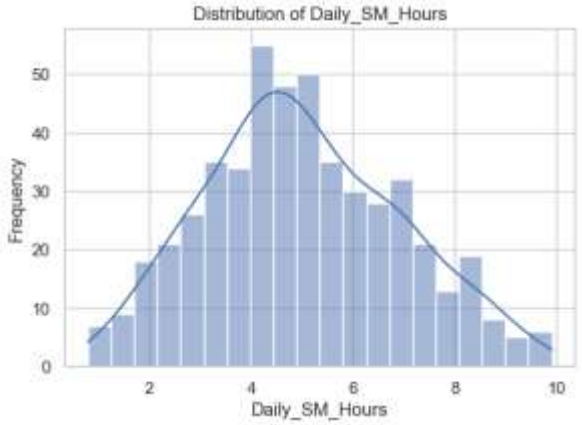
Linear regression: Study Hours is the strongest positive predictor of GPA (coef +0.126); sleep, social, extracurricular, and activity hours contribute only marginal, mostly negative effects once study time is controlled for.

Exploratory Data Analysis

Mental Health Student Dataset — Screen Time & Distress Correlations



Correlation heatmap: Daily_SM_Hours correlates strongly with Anxiety (r=0.85) and Wellbeing (r=−0.83).



Daily social media use spans ~0–8 hrs; most students cluster in the 2–5 hr "Medium" usage band (245 of 500).

500

students surveyed

0

missing values & duplicate rows found

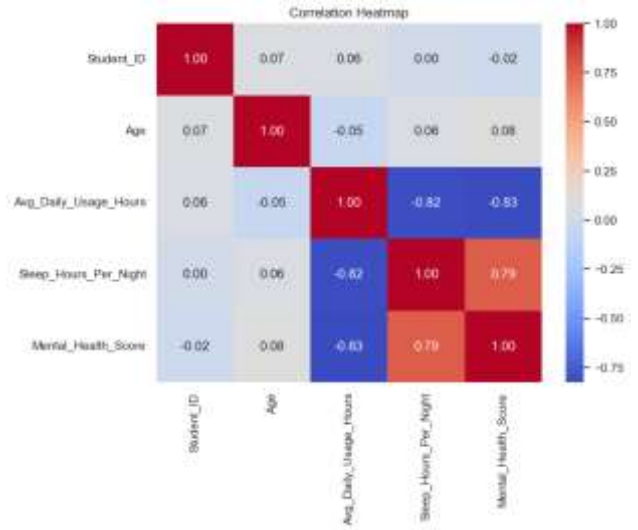
0.85

correlation between Daily_SM_Hours and Anxiety_Score — the strongest link in the dataset

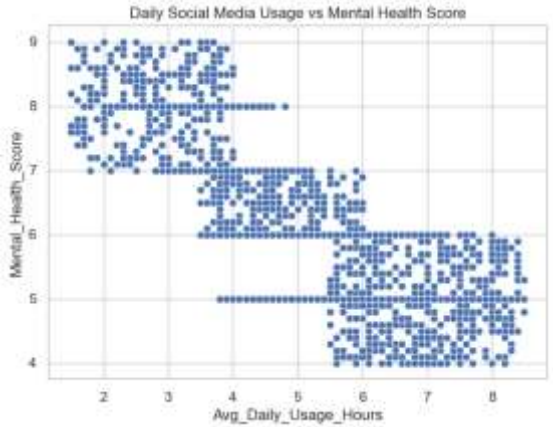
500 students, 21 variables, zero missing values or duplicates. Screen time tracks tightly with distress: SM hours also correlate with Sleep_Hours (r=−0.71), Stress_Score (r=0.81), and GPA (r=−0.78).

Exploratory Data Analysis

Social Media Impact Dataset — Usage vs. Mental Health



Correlation heatmap: heavier daily usage tracks with lower self-reported mental health scores.



Scatter of Avg_Daily_Usage_Hours vs. Mental_Health_Score shows a clear downward trend as usage climbs.

1,705

student records (1,000 unique students)

0

missing values & duplicate rows found

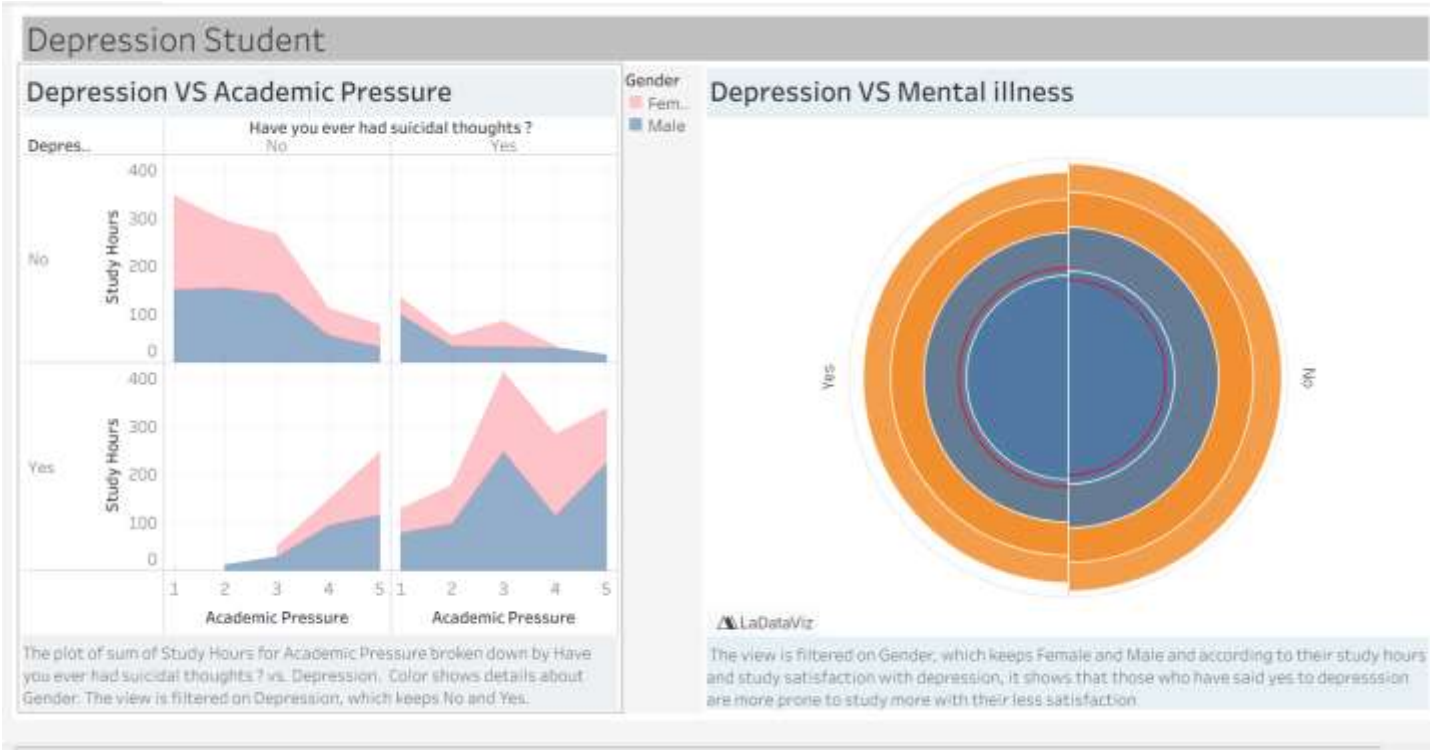
50.4%

of students are "High" users (5h+/day) — avg usage is 5.1 hrs/day

1,705 records across 1,000 students, 11 variables, zero missing values. Avg sleep is 6.6 hrs/night and avg mental health score is 6.2/10 — both trend down as daily usage rises.

Dashboard Prototype

Do suicidal thoughts and academic pressure predict depression risk?



107.3

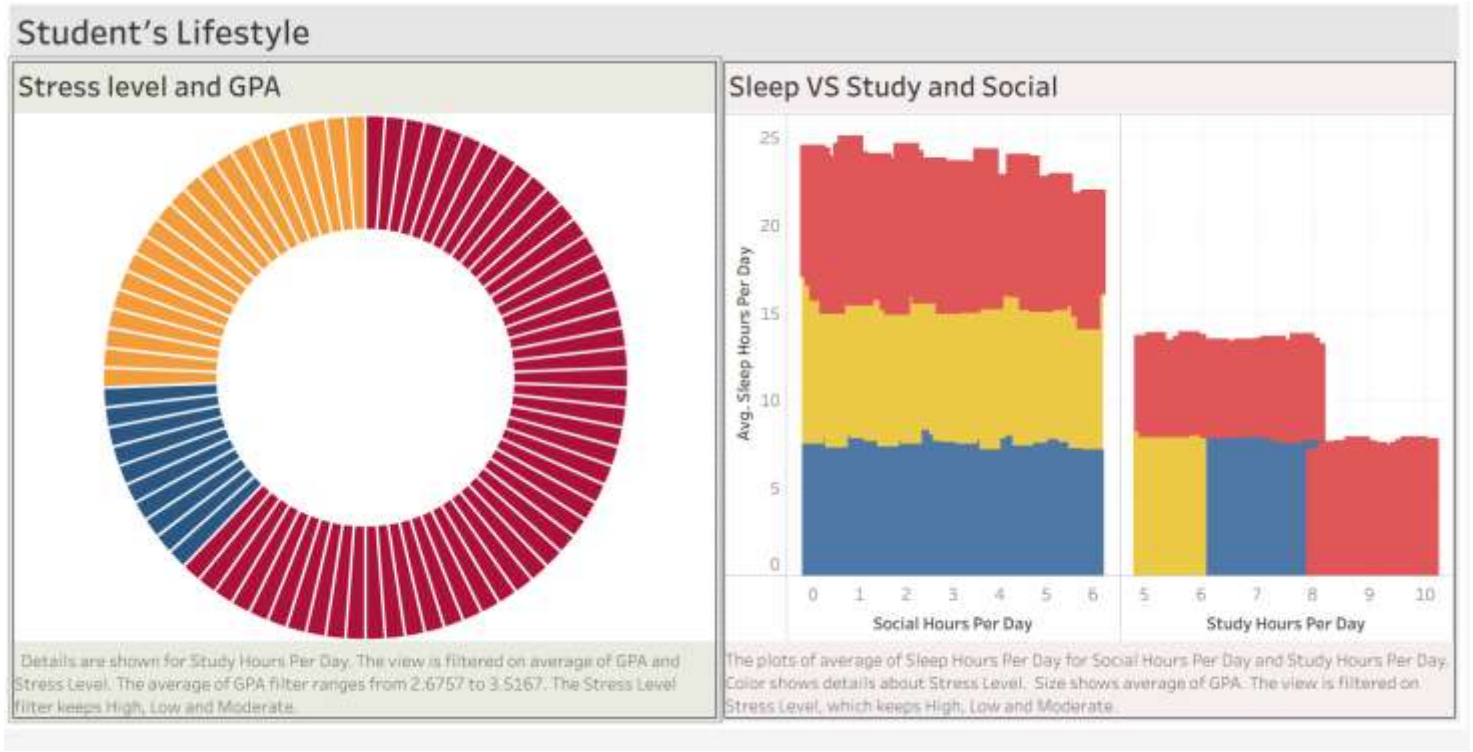
chi-square statistic linking suicidal thoughts to depression ($p < 0.001$)

Depression vs Academic Pressure & Mental Illness: students with suicidal thoughts study more as pressure rises, with lower satisfaction.

Contributor: Mahdiat Rahman

Dashboard Prototype

How do stress, sleep, study, and social time interact to shape GPA?



434.9

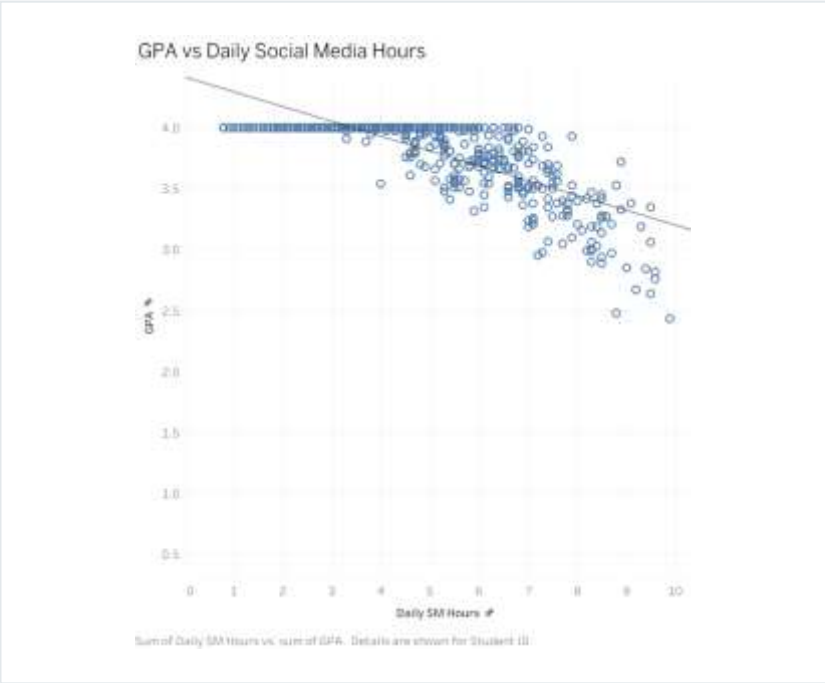
ANOVA F-statistic — GPA falls sharply as stress rises ($p < 0.001$)

Stress Level & Sleep vs Study/Social: high-stress students dominate as daily study hours rise and social hours fall.

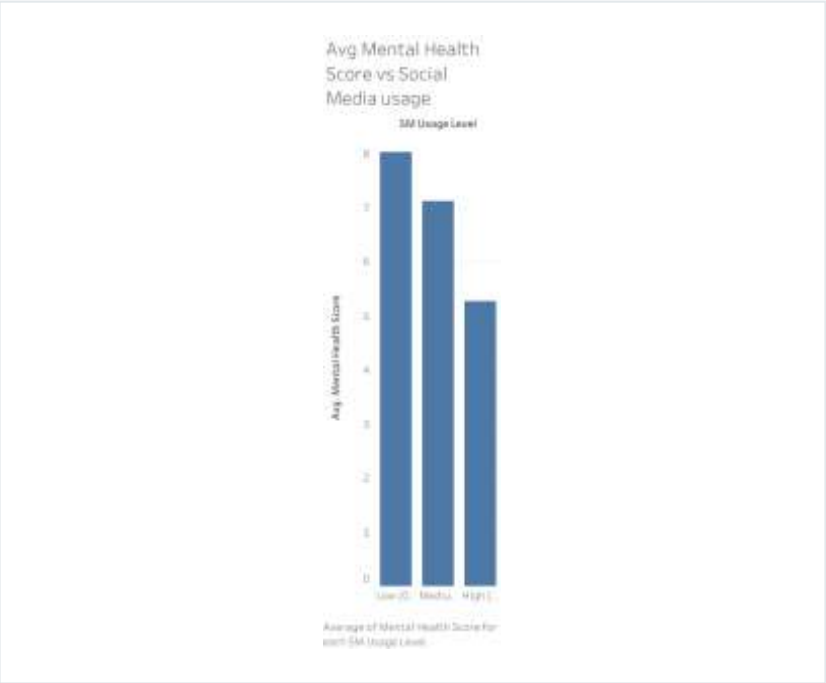
Contributor: Mahdiat Rahman

Dashboard Prototype

Does daily social media usage relate to GPA and mental health?



GPA vs Daily SM Hours: a downward trend line shows GPA falls as daily usage rises (n=500).



Avg Mental Health Score by usage tier: drops steadily from 8.03 (Low) to 5.27 (High) (n=1,705).

-2.76

point drop in avg. mental health score, Low vs High usage

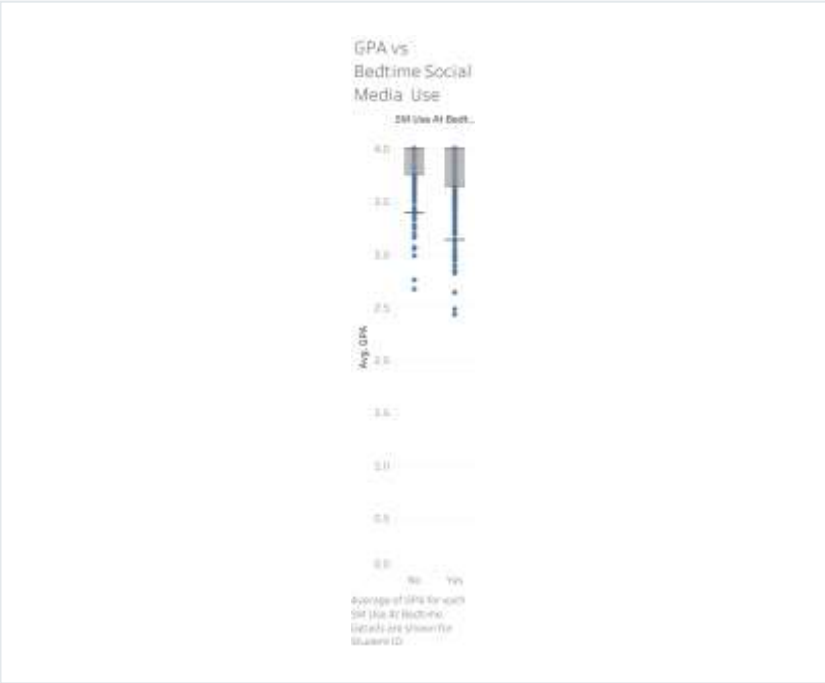
500

students show GPA declining as daily SM hours increase

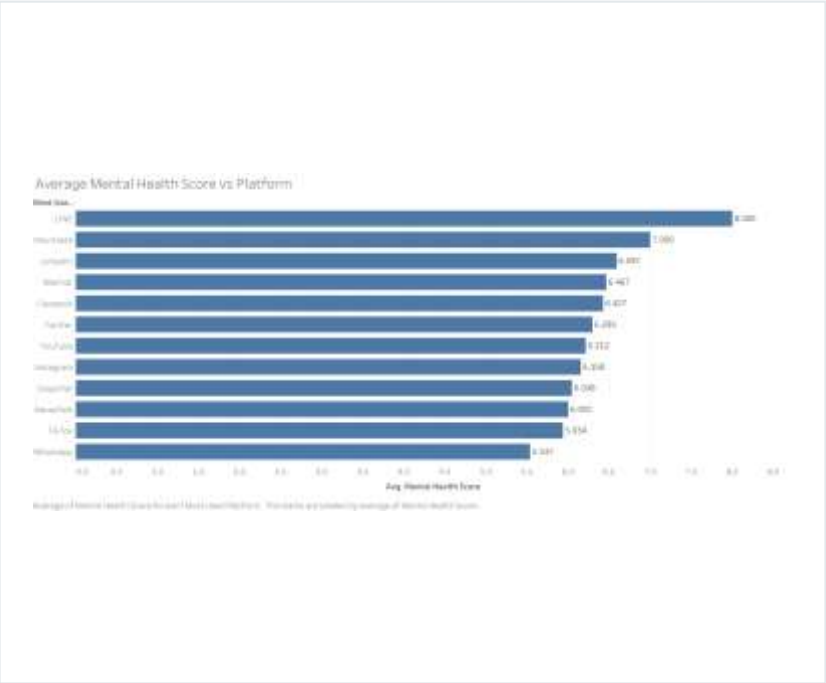
Contributor: Rabeta Tanjum Arni

Dashboard Prototype

Does timing and platform choice matter too?



GPA by Bedtime SM Use: median GPA is lower for students who use social media at bedtime (~3.15 vs ~3.4).



Avg Mental Health Score by platform: scores range from 8.0 (LINE) down to 5.5 (WhatsApp), n=1,705.

8.0

highest avg. mental health score (LINE users)

5.5

lowest avg. mental health score (WhatsApp users)

Contributor: Rabeta Tanjum Arni

Conclusion

Across all four datasets, the story is consistent: heavier screen and social media use tracks with higher stress, greater depression risk, less sleep, and lower GPA.

1. Screen time is the strongest signal in the data — daily social media hours correlate with anxiety (r = 0.85) and inversely with wellbeing (r = -0.83) and GPA (r = -0.78).

2. Depressed and high-stress students study more, sleep less, and report lower satisfaction — pressure, not idleness, is driving the outcome.

3. Study hours, not sleep or social time, is the strongest positive predictor of GPA — making digital-habit boundaries a realistic, data-backed intervention target.

Thank You

Questions & Discussion